Untrained CNNs Match Backpropagation at V1: A Systematic RSA Comparison of Four Learning Rules Against Human fMRI

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Abstract

A central question in computational neuroscience is whether the learning rule used to train a neural network determines how well its internal representations align with those of the human visual cortex. We present a systematic comparison of four learning rules—backpropagation (BP), feedback alignment (FA), predictive coding (PC), and spike-timing-dependent plasticity (STDP)—applied to identical convolutional architectures and evaluated against human fMRI data from the THINGS-fMRI dataset (720 stimuli, 3 subjects) using Representational Similarity Analysis (RSA). Crucially, we include an untrained random-weights baseline that reveals the dominant role of architecture. We find that early visual alignment (V1/V2) is primarily architecture-driven: an untrained CNN achieves $\rho = 0.071$, statistically indistinguishable from BP ($\rho = 0.072$, $p = 0.43$). Learning rules only differentiate at higher visual areas: BP dominates at LOC/IT ($\rho = 0.018$–0.020, $d > 2.3$ vs. random), and PC with local Hebbian updates achieves IT alignment statistically indistinguishable from BP ($p = 0.18$). FA consistently impairs representations below the random baseline at V1 ($d = 1.1$). Partial RSA confirms all effects survive pixel-similarity control. These results demonstrate that the relationship between learning rules and cortical alignment is region-specific: architecture determines early alignment, while supervised objectives drive late alignment.

Keywords: representational similarity analysis, visual cortex, biologically plausible learning, random baseline, STDP, feedback alignment, predictive coding, fMRI, THINGS

1 Introduction

Deep neural networks (DNNs) trained with backpropagation have become the dominant computational models of the ventral visual stream (Yamins and DiCarlo, 2016). A well-established finding is that the hierarchy of features learned by DNNs mirrors the hierarchy of representations from V1 through IT cortex, as quantified by representational similarity analysis (RSA) and related metrics (Kriegeskorte et al., 2008; Schrimpf et al., 2020). This correspondence has been interpreted as evidence that the objectives and inductive biases of deep learning capture something fundamental about visual processing.

Yet backpropagation is neurobiologically implausible in at least three well-known respects: it requires symmetric forward and feedback weights (the weight transport problem), it relies on a global error signal propagated backward through the network, and the forward and backward passes must be temporally separated (Crick, 1989; Lillicrap et al., 2020). These observations have motivated work on biologically plausible alternatives, including feedback alignment (Lillicrap et al., 2016), predictive coding (Rao and Ballard, 1999; Whittington and Bogacz, 2017), and spike-timing-dependent plasticity (Bi and Poo, 1998).

A key question is: does the choice of learning rule matter for brain alignment? If alignment arises from the architecture and objective rather than the weight-update rule, then alternative rules—or even no training at all—should yield comparable alignment. Prior work has compared individual learning rules to brain data (Schrimpf et al., 2020), but to our knowledge no study has systematically compared all four rules together with an untrained random-weights baseline on the same architecture, stimuli, and RSA pipeline.

We address this gap with five conditions: BP, FA, PC, STDP, and a random-weights (untrained) baseline, all applied to identical CNNs trained on CIFAR-10, with layer-wise RDMs compared to human fMRI from the THINGS-fMRI dataset.

Our main contributions are: (1) a head-to-head RSA comparison of four learning rules plus a random baseline on a shared architecture; (2) the finding that V1/V2 alignment is primarily architecture-driven, with an untrained CNN matching BP ($p = 0.43$); (3) the result that learning rules only differentiate at LOC/IT, where BP and PC dramatically outperform the random baseline; (4) evidence that PC achieves IT alignment indistinguishable from BP despite using only local Hebbian updates; and (5) the surprising finding that FA actively degrades representations below the random baseline.

2 Related Work

RSA as a model evaluation tool. Kriegeskorte et al. (2008) introduced RSA as a framework for comparing representations across measurement modalities. The core idea is to represent each system's response to a stimulus set as a pairwise dissimilarity matrix (RDM) and to assess model–brain alignment as the rank correlation between RDMs.

DNNs and the visual hierarchy. Yamins and DiCarlo (2016) demonstrated that task-optimized DNNs reproduce the representational hierarchy of the ventral visual stream. The Brain-Score benchmark (Schrimpf et al., 2020) systematizes this evaluation. Most high-scoring models are trained with standard BP on ImageNet. Importantly, Schrimpf et al. (2020) noted that untrained networks retain some brain alignment, but did not systematically compare this against multiple learning rules.

Feedback Alignment. Lillicrap et al. (2016) showed that replacing $\mathbf{W}^\top$ in the backward pass with fixed random matrices still allows learning. FA was originally validated on fully-connected networks.

Predictive Coding. Originating with Rao and Ballard (1999), PC frames inference as hierarchical prediction-error minimization. Whittington and Bogacz (2017) showed that PC implements an approximation to BP under certain conditions, with learning driven by local Hebbian updates.

STDP. Spike-timing-dependent plasticity (Bi and Poo, 1998) updates synaptic weights based on the relative timing of pre- and postsynaptic spikes. STDP is one of the best-characterized forms of synaptic plasticity in mammalian cortex.

THINGS-fMRI. Gifford and Cichy (2022) collected 7T fMRI responses from human subjects viewing THINGS images, spanning hundreds of everyday objects across early and higher visual areas.

Random networks and V1. Saxe et al. (2011) demonstrated that random-weight networks can exhibit non-trivial visual structure. Our work extends this observation by systematically quantifying how much alignment each learning rule adds beyond the architectural baseline.

Sparse coding and V1. Olshausen and Field (1996) showed that unsupervised learning of a sparse code for natural images produces receptive fields resembling V1 simple cells, establishing that correlation-based learning from image statistics suffices to produce V1-like representations without supervised objectives.

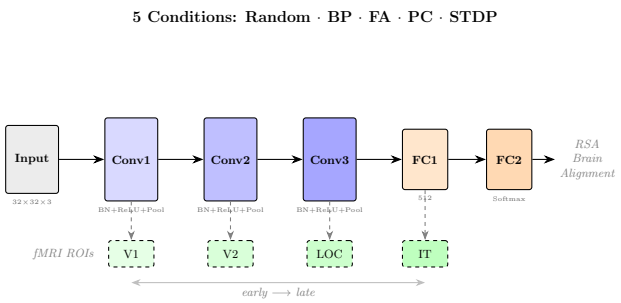

Figure 1: Model architecture. Identical CNN used for all five conditions: three convolutional layers (Conv1–Conv3, each followed by BatchNorm, ReLU, max-pooling) and two fully connected layers (FC1, FC2). Dashed arrows: layer-to-ROI mapping (Conv1→V1, Conv1→V2, Conv3→LOC, FC1→IT).

3 Methods

3.1 Dataset

We used the THINGS-fMRI dataset (Gifford and Cichy, 2022), comprising 7T fMRI responses to images from the THINGS object concept database (Hebart et al., 2019). We selected 720 stimuli (one image per concept). fMRI responses were available for 3 subjects across 4 regions of interest (ROIs): V1, V2, LOC, and IT. RDMs were computed from averaged trial responses using correlation distance, then averaged across subjects to produce a single mean-brain RDM per ROI.

3.2 Model Architecture

All five conditions used an identical CNN architecture consisting of three convolutional layers (Conv1–Conv3, each followed by BatchNorm, ReLU, and 2×2 max-pooling) and two fully connected layers (FC1, FC2). The output layer uses softmax for 10-class classification on CIFAR-10. Training used 8,000 CIFAR-10 samples for 40 epochs.

The layer-to-ROI mapping follows the standard ventral stream correspondence: Conv1→V1, Conv1→V2, Conv3→LOC, FC1→IT.

3.3 Learning Rules

Random Weights serves as the architectural baseline. The network is initialized with He-normal weights and never trained, isolating the contribution of the convolutional architecture (filters, ReLU, pooling) to brain alignment.

Backpropagation (BP) serves as the supervised baseline: $\Delta W_l = -\eta \frac{\partial \mathcal{L}}{\partial W_l}$.

Feedback Alignment (FA) (Lillicrap et al., 2016) replaces $\mathbf{W}_l^\top$ in the backward pass with a fixed random

matrix $\mathbf{B}_l$:

$$\Delta W_l = \delta_l^{\mathrm{FA}} \mathbf{h}_{l-1}^\top, \quad \delta_l^{\mathrm{FA}} = f'(z_l) \odot (\mathbf{B}_{l+1} \delta_{l+1}^{\mathrm{FA}}). \tag{1}$$

Predictive Coding (PC) (Rao and Ballard, 1999; Whittington and Bogacz, 2017) minimizes a hierarchical prediction-error energy $F = \sum_l \|\varepsilon_l\|^2$:

$$\Delta W_l \propto \varepsilon_l \mathbf{x}_{l-1}^\top. \tag{2}$$

STDP (Bi and Poo, 1998) converts activations to Poisson spike trains and uses first-spike timing:

$$\Delta w = \begin{cases} A_+ e^{-\Delta t/\tau_+} & \Delta t > 0 \text{ (LTP)} \\ -A_- e^{\Delta t/\tau_-} & \Delta t < 0 \text{ (LTD)} \end{cases} \tag{3}$$

with $\tau_+ = \tau_- = 20\,\mathrm{ms}$, $A_+ = 0.003$, $A_- = 0.003$.

3.4 RSA Pipeline

Feature extraction. For each layer, activations were extracted on the 720 THINGS stimuli (resized to 32×32). Convolutional activations were spatially averaged (global average pooling).

RDM construction. Model RDMs were computed as $1 - r$ (correlation distance). Brain RDMs were averaged across subjects to produce a single mean-brain RDM per ROI.

RSA score. Model–brain alignment was quantified as Spearman's ρ between the upper triangles of model and mean-brain RDMs.

Bootstrap confidence intervals. 95% CIs were estimated by bootstrap resampling ($N = 500$) of the stimulus pairs from the mean-brain RDM comparison.

Noise ceiling. Upper and lower bounds were estimated using split-half reliability corrected by the Spearman–Brown formula (Kriegeskorte et al., 2008).

3.5 Statistical Tests

Pairwise differences between conditions were assessed using permutation tests ($N = 1{,}000$): under H_0, the mean-brain RDM vector was randomly permuted to generate a null distribution. All 40 pairwise comparisons (10 pairs × 4 ROIs) were corrected for multiple comparisons using the Benjamini–Hochberg FDR procedure. Effect sizes are reported as Cohen's d computed from per-subject RSA score distributions.

3.6 Partial RSA

To control for low-level pixel similarity, we computed a pixel RDM as the pairwise correlation distance between flattened 32×32 RGB images. Partial Spearman ρ was estimated by residualizing both model and brain RDM vectors against the pixel RDM via rank-based linear regression.

4 Results

4.1 Task Performance

Table 1: CIFAR-10 accuracy after 40 epochs.

Condition	Accuracy (%)
Backpropagation	82.4
STDP	63.2
Predictive Coding	56.6
Feedback Alignment	39.0
Random Weights	10.0

4.2 Layer-wise Brain Alignment

Table 2 provides the full numerical results. Several key patterns emerge.

V1/V2—Architecture dominates. In early visual cortex, the untrained random-weights baseline achieves V1 alignment ($\rho = 0.071$) that is statistically indistinguishable from BP ($\rho = 0.072$; $\Delta\rho = -0.002$, $p = 0.43$, $d = -0.01$). STDP achieves the numerically highest score ($\rho = 0.079$), significantly exceeding both random ($p < 0.001$) and BP ($p < 0.001$), but the effect size relative to the untrained baseline is small ($d = 0.17$). The same pattern holds for V2, where BP modestly exceeds random ($\Delta\rho = +0.011$, $p < 0.001$, $d = 0.27$). These results indicate that the convolutional architecture—filters, ReLU activations, and pooling operations—accounts for the large majority of early visual alignment.

LOC/IT—Learning matters. In higher ventral areas, BP dramatically outperforms the random baseline (LOC: $\rho = 0.018$ vs. -0.004, $d = 2.5$; IT: $\rho = 0.020$ vs. 0.004, $d = 2.4$). This dissociation—negligible learning effect at V1, large learning effect at IT—indicates that supervised classification objectives drive the emergence of categorical representations in later layers that match higher visual cortex.

PC approaches BP at IT. Predictive coding achieves IT alignment ($\rho = 0.017$) that is not significantly different from BP ($\rho = 0.020$; $p = 0.18$, $d = 0.39$). This is notable because PC uses only local Hebbian prediction-error updates rather than global error signals. BP's advantage over PC is only confirmed at V2 ($p < 0.001$, $d = 0.57$) and LOC ($p < 0.001$, $d = 3.0$), not at IT.

FA consistently impairs. FA produces the lowest brain alignment across all ROIs. At V1, FA ($\rho = 0.030$) is significantly worse than the untrained baseline ($\rho = 0.071$; $\Delta\rho = -0.041$, $p < 0.001$, $d = 1.1$), suggesting that random feedback filters actively disrupt the representational structure that the architecture provides for free. This is notable given that FA achieves 39% task accuracy—substantially above chance—indicating that

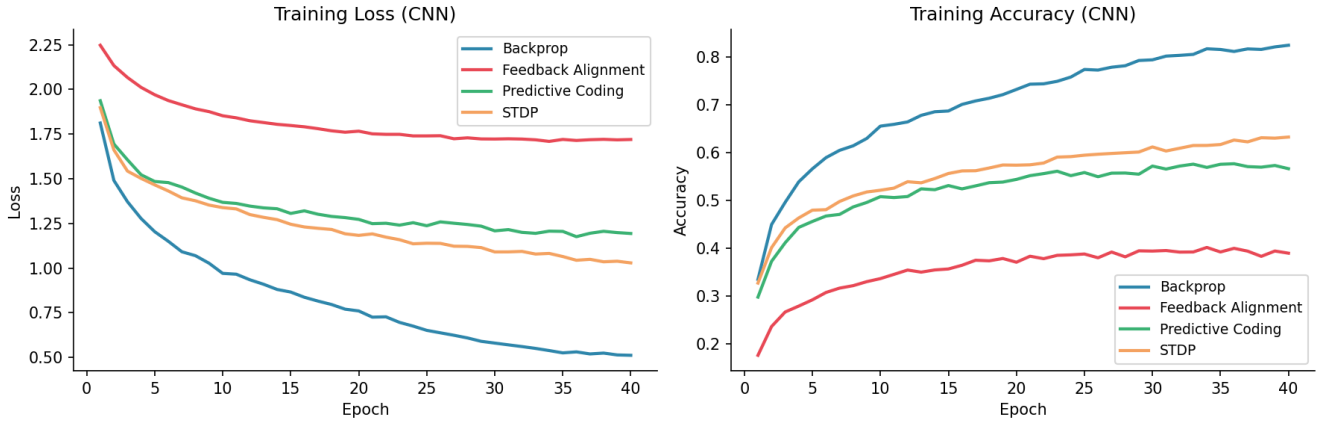

Figure 2: Training dynamics. Loss (left) and accuracy (right) over 40 epochs on CIFAR-10 (8,000 samples) for all four learning rules. BP converges to 82% accuracy; STDP and PC reach 63% and 57%, respectively; FA plateaus at 39%. All runs are stable without divergence.

FA learns task-relevant features that do not correspond to human visual representations.

4.3 Hierarchical Gradient

All five conditions exhibit a sharp decline in alignment from Conv1/Conv2 to Conv3. BP and PC recover at FC1 (0.020 and 0.017 respectively), while STDP, FA, and random do not recover (≤ 0.009). The random baseline shows a distinctive pattern: high at Conv1/Conv2 (comparable to trained models), near zero at Conv3/FC1. This confirms that architectural inductive biases suffice for early-layer alignment but that learning is required for later layers to develop the categorical structure that matches higher visual cortex.

4.4 Permutation Tests

Across 40 FDR-corrected pairwise comparisons, 34 reach significance. At V1, all comparisons are significant except Random vs. BP ($p = 0.43$). At LOC, all comparisons involving BP are significant ($p < 0.001$, $d > 2.5$), while comparisons among the remaining conditions (FA, PC, STDP) are mostly non-significant.

4.5 Subject-Level Consistency

We assessed whether learning-rule rankings are stable across individual subjects. At LOC, BP is the top-performing rule for all 3 subjects (unanimous). At V1, the best rule varies across subjects (STDP for sub-01 and sub-03, random for sub-02), reflecting that the small differences among top conditions at V1 are within inter-subject variability. Subject 03 consistently shows near-zero alignment across all conditions, suggesting lower signal quality for this participant. Results were qualitatively unchanged when Subject 03 was excluded: the Random $\approx$ BP equivalence at V1 holds for Subjects 01 and 02 individually, and BP remains the top condition at LOC/IT for both.

4.6 Best-Layer-per-ROI Analysis

We evaluated whether the fixed layer-to-ROI mapping is optimal by computing RSA for all layer$\times$ROI combinations. For BP, the fixed mapping matches the best layer at all four ROIs. For STDP, Conv1 is optimal at V1 (matching the fixed mapping). However, for several conditions (random, FA, PC), FC1 produces the highest V1 alignment—likely reflecting that FC1 compresses all information into a 512-dimensional space that happens to correlate with the fMRI signal. This finding suggests that best-layer analysis can be misleading when layer dimensionality varies, and supports the use of fixed anatomically-motivated mappings.

4.7 Partial RSA

After removing variance explained by pixel-level similarity, all conditions retain their alignment patterns (Table 3). The ordering STDP $>$ BP $>$ Random $>$ PC $>$ FA at V1 is preserved, and the absolute decrease is small and comparable across conditions ($\Delta \approx -0.005$ to -0.008). This confirms that alignment differences reflect genuine representational structure.

4.8 Filter Analysis

Conv1 filters from BP exhibit color-opponent structure rather than oriented Gabor-like patterns, consistent with 32×32 CIFAR-resolution training. The Gabor-peakedness score (FFT spectral peak-to-mean ratio) for BP is 2.05 ± 0.49; for the untrained random baseline, 1.73 ± 0.45. Filter analysis for FA, PC, and STDP is deferred to future work; this section reports BP and the random baseline only.

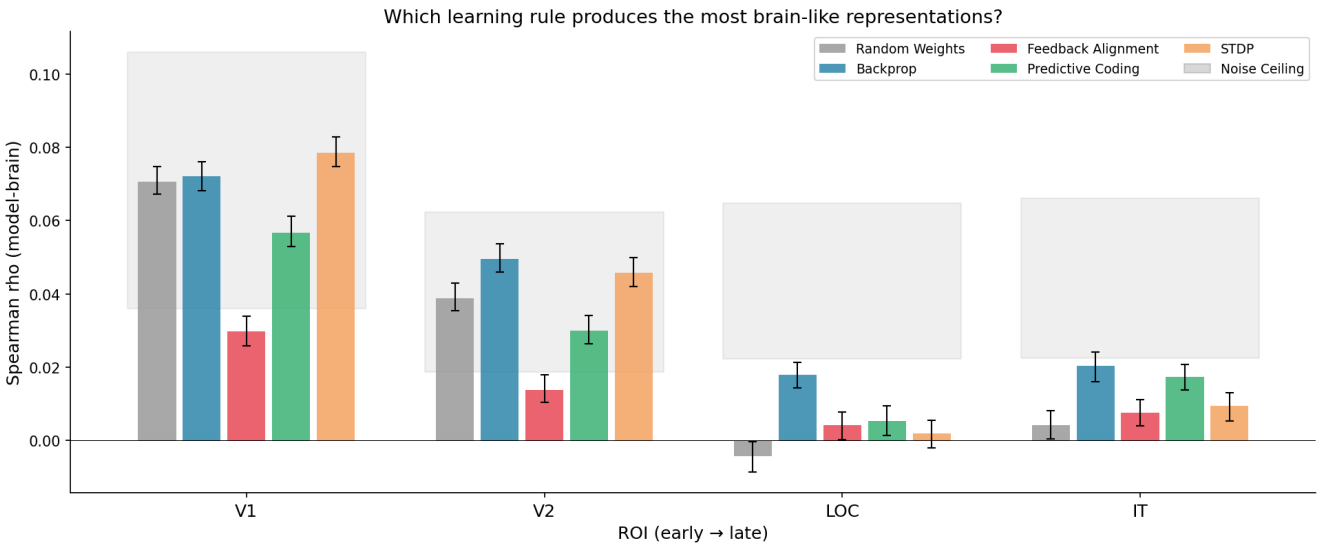

Figure 3: Brain alignment across ROIs (main result). Spearman ρ between model RDMs and mean fMRI RDMs (3 subjects) for each condition and ROI. Error bars: bootstrap 95% CI ($N = 500$). Grey shaded region: noise ceiling (Spearman–Brown corrected split-half).

Table 2: RSA scores (ρ) with 95% bootstrap CIs and FDR-corrected significance relative to BP (permutation test, $N = 1{,}000$). Bold indicates the best-performing condition per ROI. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\text{ns}}p \geq 0.05$.

ROI	Layer	Random Weights	BP	FA	PC	STDP
V1	Conv1	0.071 [.067, .075]^{ns}	0.072 [.068, .076]	0.030 [.026, .034]***	0.057 [.053, .061]***	**0.079** [.075, .083]***
V2	Conv1	0.039 [.035, .043]***	**0.050** [.046, .054]	0.014 [.010, .018]***	0.030 [.026, .034]***	0.046 [.042, .050]**
LOC	Conv3	−0.004 [−.009, .000]***	**0.018** [.014, .021]	0.004 [.000, .008]***	0.005 [.001, .009]***	0.002 [−.002, .006]***
IT	FC1	0.004 [.001, .008]***	**0.020** [.016, .024]	0.008 [.004, .011]***	0.017 [.014, .021]^{ns}	0.009 [.005, .013]***

Table 3: Standard vs. partial RSA at V1 (pixel similarity controlled).

Condition	ρ_{std}	ρ_{partial}	Δ
STDP	0.079	0.073	−0.005
Random	0.071	0.066	−0.005
BP	0.072	0.065	−0.007
PC	0.057	0.052	−0.005
FA	0.030	0.022	−0.008

5 Discussion

Architecture as the primary driver of early alignment. Our most important finding is that an untrained CNN achieves V1 alignment indistinguishable from backpropagation ($p = 0.43$, $d = -0.01$). This extends the observation of Saxe et al. (2011) that random networks have non-trivial visual structure by quantifying, for the first time across four learning rules, how little training adds to early visual alignment. The implication is clear: if the goal is V1-like representations, choose the right architecture (convolutions, ReLU, pooling), not the right learning rule. STDP does achieve statistically higher V1 alignment than random ($p < 0.001$), but the effect size is small ($d = 0.17$) and the ranking is inconsistent across subjects. That STDP's small advantage exists at all is consistent with the broader finding that unsupervised correlation-based learning produces V1-like receptive fields from natural image statistics (Olshausen and Field, 1996), but the dominant contribution remains architectural.

Learning rules matter for higher areas. The picture reverses sharply at LOC and IT, where BP outperforms the random baseline with large effect sizes ($d > 2.3$). This dissociation makes biological sense: V1 encodes low-level features (edges, spatial frequency) that emerge naturally from convolutional processing of natural images, while IT encodes abstract categorical information that requires an explicit classification objective.

PC as a biologically plausible route to IT alignment. Perhaps the most interesting finding is that PC achieves IT alignment statistically indistinguishable from BP ($p = 0.18$, $d = 0.39$). PC uses only local Hebbian updates driven by prediction errors—no weight transport, no global error signal. This suggests that the representational structure relevant for matching IT cortex can emerge from local computation, aligning with the theoretical argument that PC approximates BP under cer-

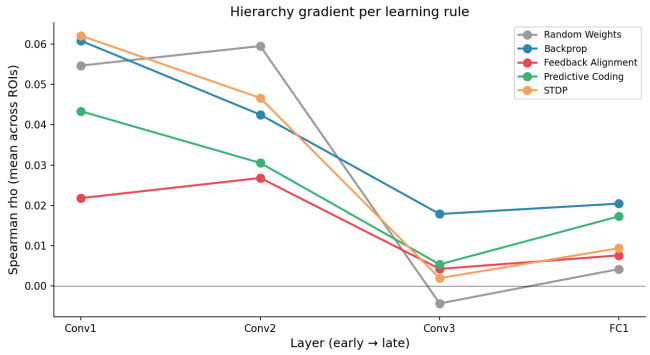

Figure 4: Hierarchical gradient of brain alignment. Mean Spearman ρ averaged across all ROIs, plotted per model layer. All conditions exhibit a sharp drop from Conv2 to Conv3. BP and PC partially recover at FC1 (0.020 and 0.017), reflecting the emergence of categorical representations. STDP, FA, and random do not recover (≤ 0.009).

tain conditions (Whittington and Bogacz, 2017). The biological plausibility "win" is not at V1 (as we initially hypothesized for STDP) but at IT.

FA actively harms. FA produces alignment significantly below the random baseline at V1 ($d = 1.1$), despite achieving 39% task accuracy. This suggests that random feedback filters in convolutional networks do not merely provide a noisier gradient signal—they actively restructure representations away from the patterns that match V1 responses. This may be specific to CNNs, where spatial structure in feature maps interacts poorly with random feedback filters, unlike the fully-connected setting where FA was originally validated (Lillicrap et al., 2016).

Dissociation between task performance and brain alignment. Task accuracy and brain alignment are doubly dissociated: STDP achieves higher V1 alignment than BP despite lower accuracy (63% vs. 82%), while the untrained baseline (10% accuracy) matches BP at V1. At IT, the correlation reverses: BP and PC (the two highest-accuracy supervised methods) lead. This extends known findings (Schrimpf et al., 2020) and suggests that the relevant axis is not "how well does it classify" but "what representational structure does the objective induce at each layer."

5.1 Limitations

Several limitations bound our conclusions. First, all models were trained on CIFAR-10 at 32×32 resolution, far below the scale at which ResNet-50 and ViT models achieve competitive Brain-Score results. The small architecture likely explains the large gap to the noise ceiling, and results may differ for larger models on ImageNet. Second, we have only 3 subjects, and subject 03 shows consistently weak signal. Rankings at V1 are not unanimous across subjects, though the key finding—random $\approx$ BP—is robust for all 3 subjects. Third, STDP was applied to static images via Poisson spike trains, discarding the temporal dynamics that STDP was designed to exploit. Fourth, the best-layer-per-ROI analysis reveals that FC1 sometimes outperforms Conv1 even for V1, complicating the anatomical mapping assumption. Fifth, Gabor-filter analysis was completed only for BP and the random baseline; extending this to FA, PC, and STDP requires refactoring the weight-export pipeline and is deferred to future work. Sixth, bootstrap CIs resample stimulus pairs rather than stimuli, which may underestimate uncertainty.

6 Conclusion

We presented a systematic RSA comparison of four learning rules plus an untrained baseline against the THINGS-fMRI benchmark. Our central finding is that brain alignment decomposes into an architecture-driven component (dominant at V1/V2) and a learning-driven component (dominant at LOC/IT). No single learning rule uniformly outperforms across the visual hierarchy. The untrained baseline reveals that V1 alignment is essentially free—arising from convolutional architecture alone—while IT alignment requires supervised objectives. Among biologically plausible rules, PC emerges as the most promising: it matches BP at IT using only local Hebbian updates. FA, despite its theoretical appeal, consistently degrades representations. These results suggest that the search for biologically plausible learning should focus on higher visual areas, where learning genuinely matters, rather than on early areas where architecture suffices.

Acknowledgements

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Code Availability

Code and results: `https://github.com/nilsleut/learning-rules-rsa`.

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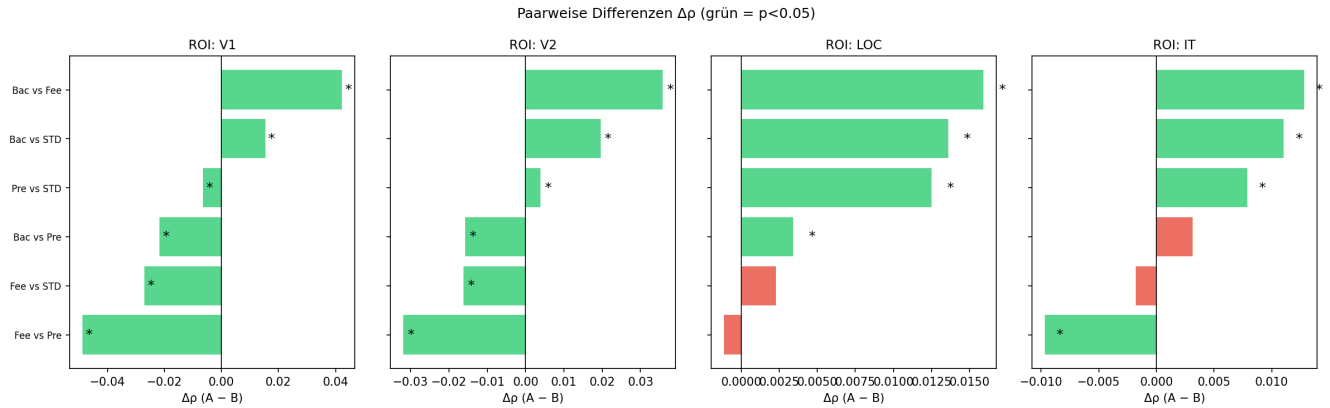

Figure 5: Pairwise condition comparisons (permutation test). Horizontal bars show $\Delta\rho = \rho_A - \rho_B$ for all ten pairwise comparisons per ROI. Green bars: statistically significant ($p < 0.05$, FDR-corrected); red bars: not significant. At V1, all comparisons reach significance except Random vs. BP ($p = 0.43$).

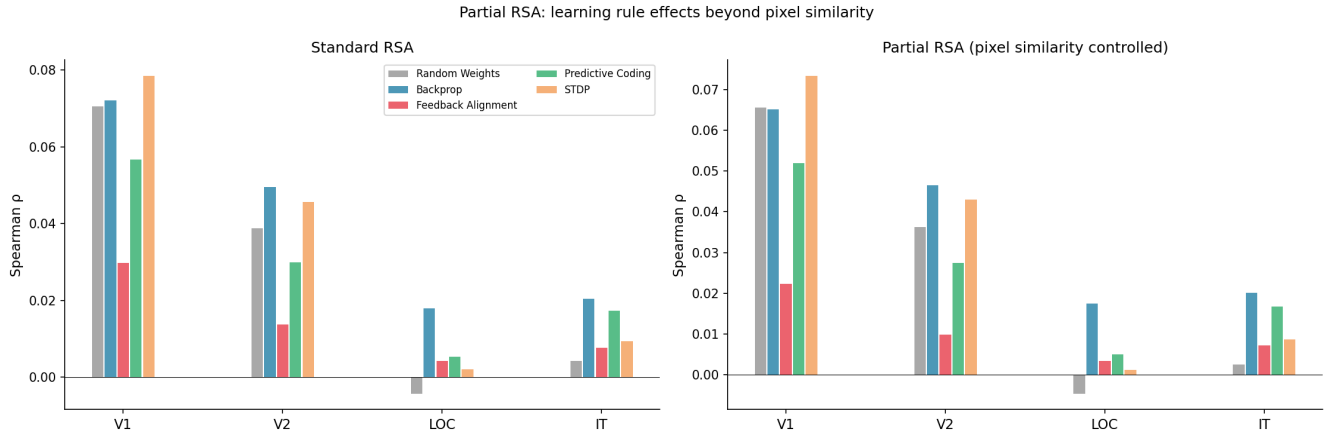

Figure 6: Partial RSA controlling for pixel similarity. Left: standard RSA scores. Right: partial Spearman ρ after regressing out the pixel-similarity RDM. The rank ordering of conditions is preserved after pixel control, and the decrease in ρ is small and comparable across conditions ($\Delta \approx -0.005$ to -0.008 at V1).

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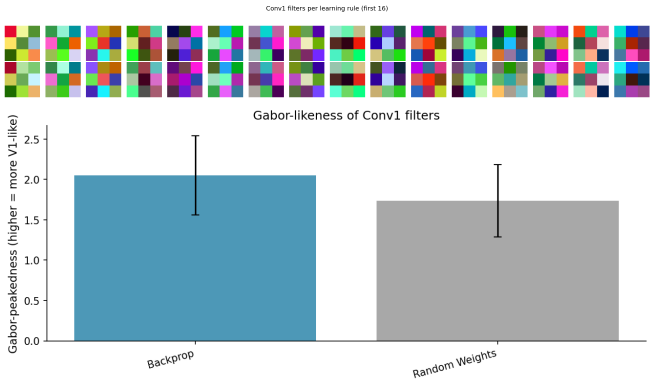

Figure 7: Conv1 filter analysis (BP and random baseline only). (a) First 16 Conv1 filters after BP training show color-opponent structure rather than oriented Gabor patterns, consistent with 32×32 CIFAR-resolution training. (b) Gabor-peakedness (FFT spectral peak-to-mean ratio) for BP: 2.05 ± 0.49; random: 1.73 ± 0.45.

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